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Boeing 747 Wing Analysis

- Problem 4:

Here, 2 cases are being analyzed. One is when the wing is taking off, and the other is when the wing is about to leave the plane. The analysis consists of treating the wings as a hollow rectangular beam attached to the plane's fuselage. The necessary loads and calculations presented in the shared PDF demonstrate the results per individual wing. The fatigue analysis where only performed for the vertical loads of the wing.

The figures below show the singularity functions of the wing taking off and their corresponding diagrams:

```
q=1353373.243*sinfun(x,0,-2)-746192.745*sinfun(x,0,-1)+127713.838*sinfun(x,0,
0)-4126.457*sinfun(x,0,1)-36182.735*sinfun(x,0,0)+36182.735*sinfun(x,30.95,0)
-55161.63*sinfun(x,10.32,-1)-55161.63*sinfun(x,20.63,-1); %q or w!

V=1353373.243*sinfun(x,0,-1)-746192.745*sinfun(x,0,0)+127713.838*sinfun(x,0,1
)-4126.457*1/2*sinfun(x,0,2)-36182.735*sinfun(x,0,1)+36182.735*sinfun(x,30.95
,1)-55161.63*sinfun(x,10.32,0)-55161.63*sinfun(x,20.63,0);

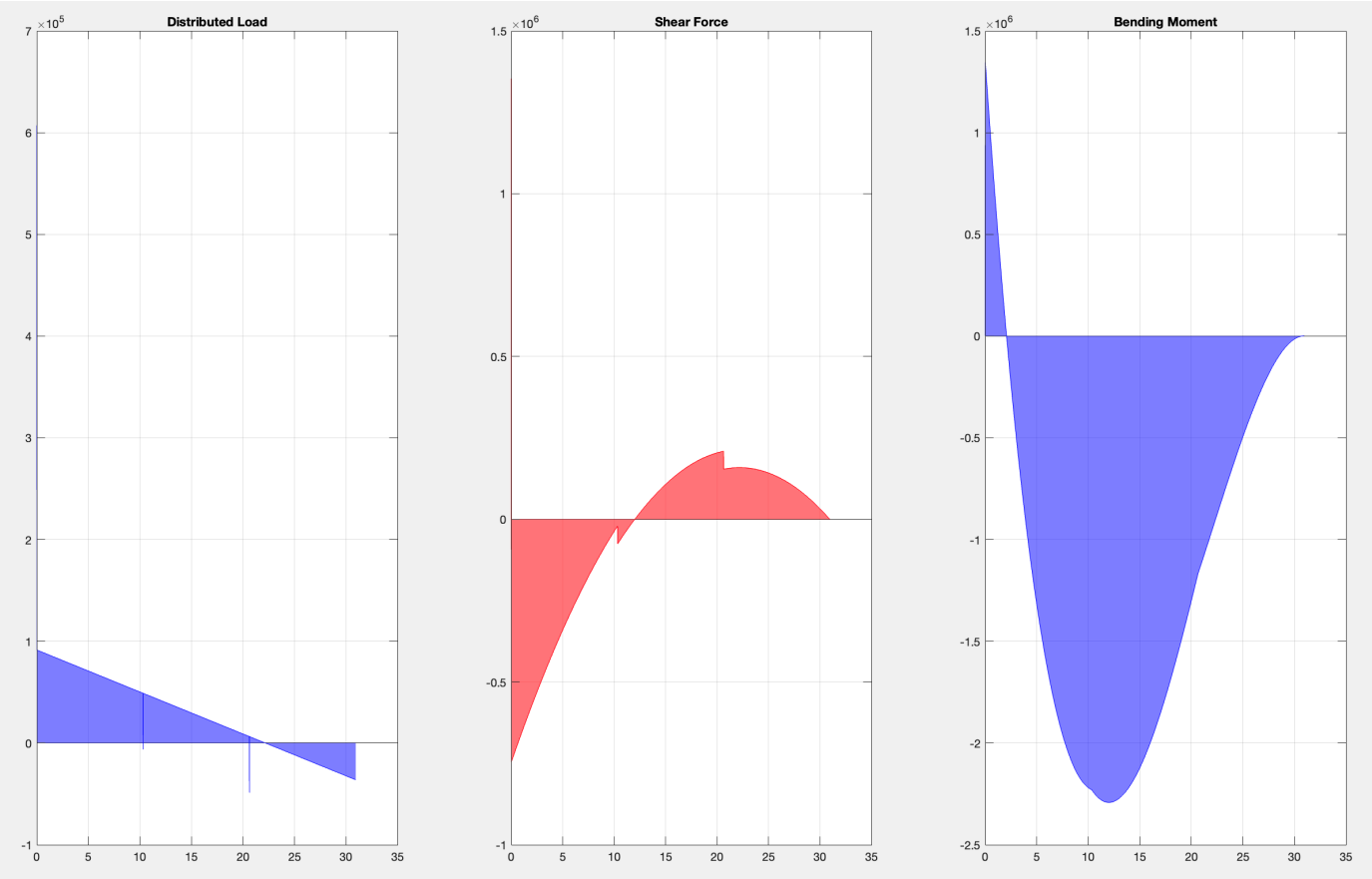
M=1353373.243*sinfun(x,0,0)-746192.745*sinfun(x,0,1)+127713.838*1/2*sinfun(x,
0,2)-1/3*1/2*4126.457*sinfun(x,0,3)-1/2*36182.735*sinfun(x,0,2)+1/2*36182.735
*sinfun(x,30.95,2)-55161.63*sinfun(x,10.32,1)-55161.63*sinfun(x,20.63,1);

rot=1/
(E*In)*(1353373.243*sinfun(x,0,1)-746192.745*1/2*sinfun(x,0,2)+127713.838*1/2
*1/3*sinfun(x,0,3)-1/3*1/2*1/4*4126.457*sinfun(x,0,4)-1/2*1/3*36182.735*sinfu
n(x,0,3)+1/2*1/3*36182.735*sinfun(x,30.95,3)-55161.63*1/2*sinfun(x,10.32,2)-5
5161.63*1/2*sinfun(x,20.63,2)+C1);

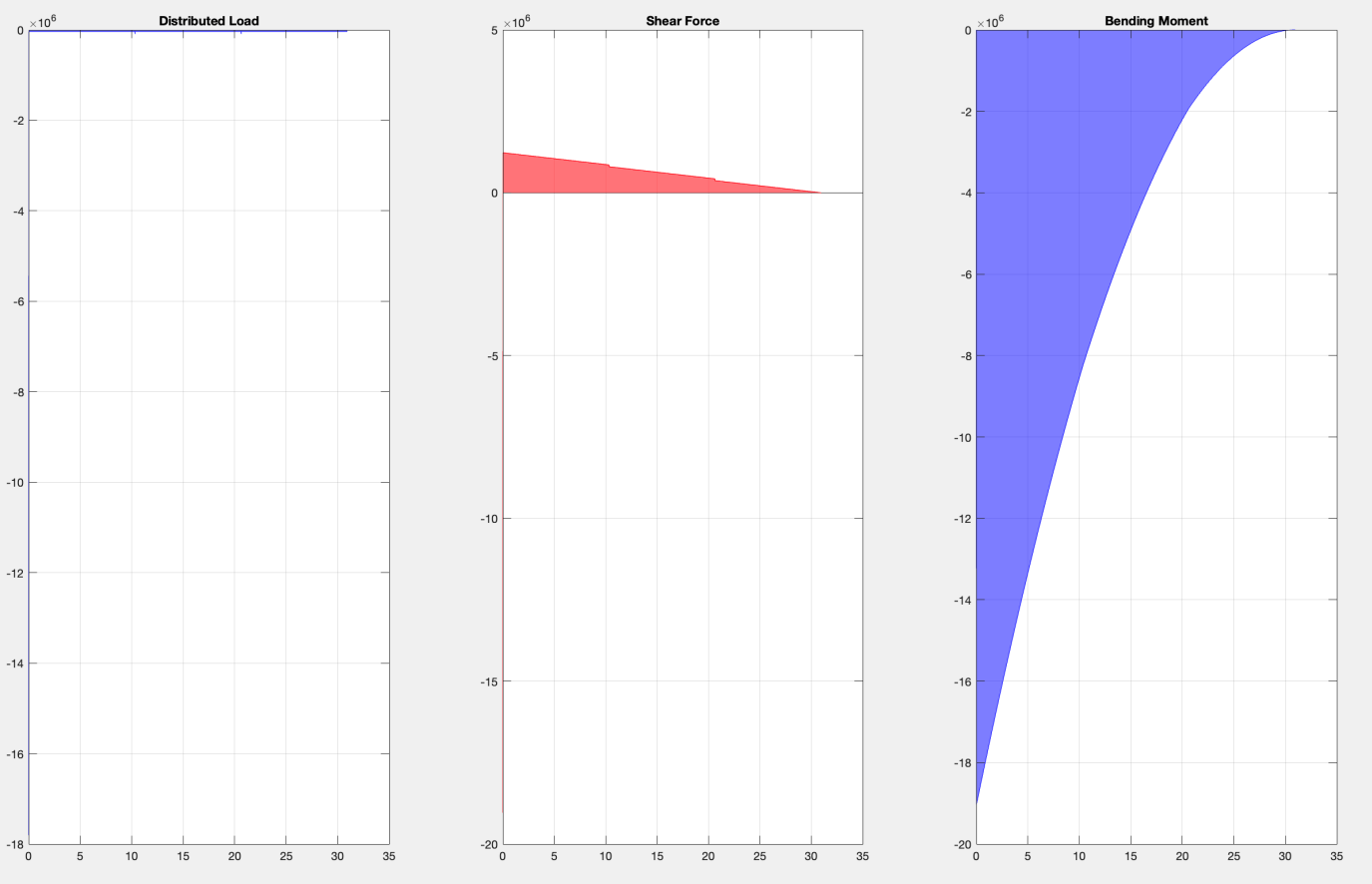
def=1/E/
In*(1353373.243*1/2*sinfun(x,0,2)-746192.745*1/2*1/3*sinfun(x,0,3)+127713.838
*1/2*1/3*1/4*sinfun(x,0,4)-1/3*1/2*1/4*1/5*4126.457*sinfun(x,0,5)-1/2*1/3*1/4
*36182.735*sinfun(x,0,4)+1/2*1/4*1/3*36182.735*sinfun(x,30.95,4)-55161.63*1/2
*1/3*sinfun(x,10.32,3)-55161.63*1/2*1/3*sinfun(x,20.63,3)+C1*x+C2);

C1=1.9606e+07, C2=0
```

The attached PDF includes the loads and calculations. The material selected for this analysis was aluminum 110 sheet, annealed. The failure analysis was conducted on the top and bottom surfaces of the beam where the wing attaches to the fuselage.



The figures below show the singularity functions of the wing taking off and their corresponding diagrams:



The singularity functions of the stationary wing are shown below:

```
q=-19036853.07*sinfun(x,0,-2)+1230178.905*sinfun(x,0,-1)-36182.735*sinfun(x,0
,0)+36182.735*sinfun(x,30.95,0)-55161.63*sinfun(x,10.32,-1)-55161.63*sinfun(x
,20.63,-1)

V=-19036853.07*sinfun(x,0,-1)+1230178.905*sinfun(x,0,0)-36182.735*sinfun(x,0,
1)+36182.735*sinfun(x,30.95,1)-55161.63*sinfun(x,10.32,0)-55161.63*sinfun(x,2
0.63,0);

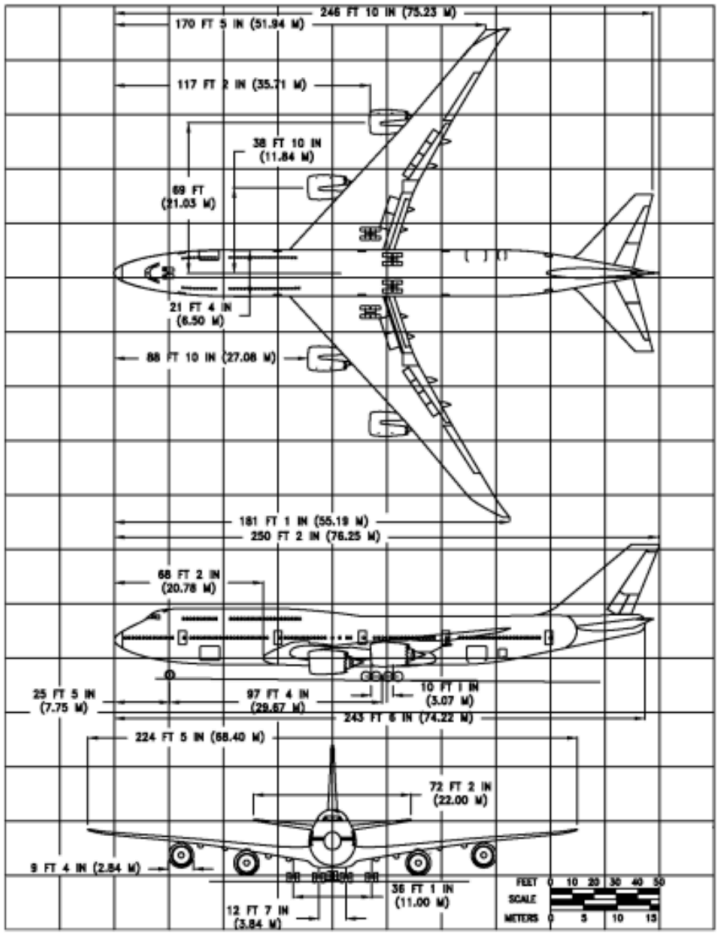
M=-19036853.07*sinfun(x,0,0)+1230178.905*sinfun(x,0,1)-1/2*36182.735*sinfun(x
,0,2)+1/2*36182.735*sinfun(x,30.95,2)-55161.63*sinfun(x,10.32,1)-55161.63*sin
fun(x,20.63,1)

rot=1/
(E*In)*(-19036853.07*sinfun(x,0,1)+1230178.905*1/2*sinfun(x,0,2)-1/2*1/3*3618
2.735*sinfun(x,0,3)+1/2*1/3*36182.735*sinfun(x,30.95,3)-55161.63*1/2*sinfun(x
,10.32,2)-55161.63*1/2*sinfun(x,20.63,2)+C1)

def=1/E/
In*(-19036853.07*1/2*sinfun(x,0,2)+1230178.905*1/2*1/3*sinfun(x,0,3)-1/2*1/3*
1/4*36182.735*sinfun(x,0,4)+1/2*1/4*1/3*36182.735*sinfun(x,30.95,4)-55161.63*
1/2*1/3*sinfun(x,10.32,3)-55161.63*1/2*1/3*sinfun(x,20.63,3)+C1*x+C2)

C1=1.9616e+07, C2=0
```

2.2.2 General Dimensions: Model 747-8



- References:
- https://www.boeing-747.com/boeing_747_family/747-8i.php
- https://www.aircraftinvestigation.info/airplanes Boeing_747-8I.html#:~:text=operating%20empty%20weight%20:%20220128%20%5Bkg,alternate)
- https://www.flightglobal.com/in-focus-boeing-747-8-technical-description-and-cutaway/107801.article
- https://www.boeing-747.com/boeing_747_family/747-8i.php • https://www.mide.com/air-pressure-at-altitude-calculator